HPC Lab 01: Performance Profiling and Optimization Report

1. Overview

This report analyzes the performance of Bubble Sort (baseline) and Quick Sort (optimized) when processing 18,250 daily temperature values (representing ~50 years of data). Performance metrics were captured using Linux profiling tools across three compiler optimization levels (`-O0,O2,O3`).

2. Timing Metrics & Performance Comparison

Baseline Execution Times (`time ./mysort`)

- **Real Time:** 0m0.789s
- **User Time:** 0m0.788s
- **System Time:** 0m0.001s

Hardware & Profiling Performance Comparison Matrix

Parameter	Bubble Sort	Quick Sort
Execution Time	0.786884 s	0.002999 s
CPU Cycles	2,531,103,344	Lower than Bubble Sort
Instructions	5,664,119,347	Lower than Bubble Sort
IPC	2.24	Higher efficiency
Cache References	12,965,773	Lower than Bubble Sort
Cache Misses	1,085,343	Lower than Bubble Sort
Hotspot Function (GPROF)	bubbleSort() (100% CPU time)	-

3. Performance Analysis & Questions

Question 1: Which sorting algorithm performed better? Explain your observation.

Quick Sort performed much better than Bubble Sort. It sorted the same dataset in about 0.002–0.005 seconds, while Bubble Sort took about 0.78 seconds.

Question 2: Which function consumed the maximum execution time?

The GPROF report shows that bubbleSort consumed 100% of the execution time. This makes it the main performance bottleneck in the program.

Question 3: How does execution time change with increasing input size?

As the number of input elements increases, the running time of both algorithms increases. However, Bubble Sort takes much longer because it repeatedly compares adjacent elements, while Quick Sort handles larger inputs more efficiently.

Question 4: How did compiler optimizations (-O2 and -O3) improve performance?

The -O2 and -O3 optimizations reduced Quick Sort's execution time by generating more efficient code and removing unnecessary overhead. Bubble Sort showed little improvement because its nested loop structure limits optimization benefits.

Question 5: Which algorithm would you recommend for large datasets? Justify your answer.

I would recommend Quick Sort because it executes much faster and handles larger inputs more efficiently than Bubble Sort.